Volume time t^* is expressed as:

$$t^* = \frac{V_t}{ADV} \tag{7.7}$$

Trading time can also be written in terms of trade rate α and POV rate. This calculation is as follows. Suppose the order is comprised of X shares. Then we can write trading time as:

$$t^* = \frac{V_t}{ADV} \cdot \left(\frac{X}{X}\right) = \frac{X}{ADV} \cdot \frac{V_t}{X}$$

In terms of trade rate α we have:

$$t^* = \frac{X}{ADV} \cdot \alpha^{-1} \tag{7.8}$$

In terms of POV rate we have:

$$t^* = \frac{X}{ADV} \cdot \frac{1 - POV}{POV} \tag{7.9}$$

TRADING RISK COMPONENTS

The timing risk (TR) measure is a proxy for the total uncertainty surrounding the cost estimate. In other words, it is the standard error of our forecast. This uncertainty is comprised of three components: price uncertainty, volume variance, and parameter estimation error. These are further described as follows:

Price Volatility: price volatility refers to the uncertainty surrounding price movement over the trading period. It will cause trading cost (ex-post) to be either higher or lower depending upon the movement and side of the order. For example, if the price moves up $0.50/share, this movement results in a higher cost for buy orders but a lower cost (savings) for sell orders. For a basket of stock, price volatility also includes the covariance or correlation across all names in the basket. Price volatility is the most commonly quoted standard error for market impact analysis. It is also very often the only standard error component.

Volume Variance: volume variance refers to the uncertainty in volumes and volume profiles over the trading horizon which could be less than, equal to, or more than a day. For example, if an investor trades an order over the full day, the cost will be different if total volume is 1,000,000 shares, 5,000,000 shares, or only 200,000 shares.